

Discovered open port 53/udp on 192.168.1.1	
Initiating Connect Scan at 21:57	
Scanning 192.168.1.1 [1000 ports] Discovered open port 53/tcp on 192.168.1.1 Discovered open port 800/tcp on 192.168.1.1	Only two open ports found.
Nmap scan report for 192.168.1.1 Host is up (0.043s latency). Not shown: 999 open/filtered ports, 998 filtered ports	
PORT STATE SERVICE 53/tcp open domain 800/tcp open mdbus_daemon 53/udp open domain	Service discovery gives more details
MAC Address: 00:21:97:88:28:21 (Elitegroup Computer System)	
Nmap done: 1 IP address (1 host up) scanned in 37.78 seconds	

The scanning continues further for service detection:

```
nmap -vv -n -sV -p:800 192.168.1.1
```

This gives the following details:

```
PORT      STATE      SERVICE      VERSION
800/tcp    open       http-proxy   Squid webproxy 2.7.STABLE9
```

Scanning all ports on the firewall is done by issuing the following command:

```
nmap -vv -sU -sT -p T:1-65535,U:1-65535 -n -r -T4  
-oNmapIPCopInternalAllPorts.txt 192.168.1.1
```

This single command scans all TCP and UDP ports in one go. The results are the same as for the earlier scan—only two open ports. The UDP scan requires more than three minutes, while the TCP scan requires about 100 seconds.

Observations

After scanning all TCP and UDP ports, the result clearly shows that the only open ports are port 53 (offering domain service) and TCP port 800 (offering Squid webproxy 2.7.STABLE9 service). All other ports are closed. Even the NTP service has been stopped, and the corresponding port



CAUTION!

While using Nmap, be warned that under the stringent rules of the Indian Cyber Law 2000, and its further amendments till date, even a port scan on a public IP could land you in jail. Do not scan computers that you do not own; and do not run scans over networks that you do not own, without written permission from the owners.

You may scan scanme.nmap.org for testing/learning. This permission only includes scanning via Nmap; you are not allowed to test other exploits and/or denial-of-service attacks. Don't forget to follow the rules listed in the Nmap man page. Abuse of this service will be reported to the government by the site owners. You have been warned.

Use Nmap very carefully, and only for discovery/audits of your own network. As mentioned earlier, it is a powerful tool, and could cause disruption/damage to the target system or network. Fully understand what you are doing, especially with scripted scans, before you do it. Ignorance will not excuse anybody from being prosecuted under the law.

Also note that if you have a website of your own, either hosted at a hosting provider, or on a rented physical server, the server and network do NOT belong to you even though you own the website's content. You should ideally obtain permission from such hosting providers/server owners to carry out even 'testing' probes of your own website.

123 is closed, instead of offering the service to the internal network. Also, note that port 80, which was being used for transparent proxy, is closed.

Further recommendations for this firewall

1. Schedule regular scans for the firewall from inside as well as from outside, to check rule modifications/changes.
2. Monitor firewall and proxy logs regularly.
3. Check for IPCop version updates/security patches regularly. 

References and further reading

- [1] <http://www.nmap.org>
- [2] <http://www.seclist.org>
- [3] http://diaryproducts.net/about/operating_systems/unix/nmap_port_scanner_iptables_firewall

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